CSE 203B W26 Homework 1

Due Time : 11:50pm, Friday Jan. 16, 2026 Submit to Gradescope
Gradescope: https://gradescope.com/

In this homework, we work on the basic concepts of convex optimization and linear algebra. You are free to use software tools to facilitate the calculation. However, you are also encouraged to go through the derivations manually wherever it is feasible because these concepts are fundamental and needed for the study of convex optimization.

All the problems are graded by completion with demonstration of reasonable efforts.

Homeworks are to be submitted by each group of up to four members with a description of the contributions of each member.

<u>IMPORTANT:</u> Mention the contributions of all members towards the homework in the beginning of the report.

1. Convex Optimization (10 pts)

1.1. Given a function $f_0(x) = 3x^4 - 4x^3 + 6x^2 - 12x - 5$, where $x \in \mathbb{R}$. Solve $\min_x f_0(x)$ using Karush-Kuhn-Tucker conditions. Show your derivation. (5 pts)

1.2. Given two functions $f_0(x) = x^2 - 2x + 2$, and $f_1(x) = 3x - 2$, where $x \in \mathbb{R}$. Solve $\min_x f_0(x)$ subject to $f_1(x) \leq 0$. (5 pts)

2. Matrix Properties (16 pts)

2.1. Linear System:

Consider the following system of linear equations

$$\begin{aligned} x_1 + x_2 \qquad\quad &= 1 \\ x_1 + 2x_2 + 3x_3 &= -3 \\ x_2 + 3x_3 &= 2. \end{aligned}$$

Write the equations in a matrix form. (2 pts)

2.2. For the matrix in problem 2.1, derive its range. What's the rank of this matrix? (2pts)

2.3. Derive the nullspace of the matrix in problem 2.1. What's the relation between the range and nullspace of a matrix? (2pts)

2.4. Derive the trace and determinant of the matrix in problem 2.1. Write the eigenvalues and eigenvectors. (3pts)

2.5. Prove the following properties. Note that det means determinant operator and tr trace opeartor. (3 pts)

- For $A, B \in \mathbb{R}^{n\times n}$, $\det AB = \det A \times \det B$.
- For $A \in \mathbb{R}^{m\times n}, B \in \mathbb{R}^{n\times m}$, $\operatorname{tr} AB = \operatorname{tr} BA$.
- For $A \in \mathbb{R}^{n\times n}$, $\det A = \prod_{i=1}^{n} \lambda_i$, and $\operatorname{tr} A = \sum_{i=1}^{n} \lambda_i$, where $\lambda_i, i = 1, \ldots, n$ are the eigenvalues of A.

2.6. Use the following example of matrices A and B to illustrate the equations in problem 2.5. (4 pts)

$$A = \begin{bmatrix} 4 & -1 & -1 \\ -1 & 4 & -1 \\ -1 & -1 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix}.$$

3. Matrix Operations (24 pts)

Gradient: consider a function $f : \mathbb{R}^n \to \mathbb{R}$ that takes a vector $x \in \mathbb{R}^n$ and returns a real value. Then the gradient of f (w.r.t. x) is the vector of partial derivatives, defined as

$$\nabla_x f(x) = \begin{bmatrix} \frac{\partial f(x)}{\partial x_1} \\ \frac{\partial f(x)}{\partial x_2} \\ \vdots \\ \frac{\partial f(x)}{\partial x_n} \end{bmatrix}.$$

Hessian: consider a function $f : \mathbb{R}^n \to \mathbb{R}$ that takes a vector $x \in \mathbb{R}^n$ and returns a real value. Then the Hessian matrix of f (w.r.t. x) is the $n \times n$ matrix of partial derivatives, defined as

$$\nabla_x^2 f(x) = \begin{bmatrix} \frac{\partial^2 f(x)}{\partial x_1^2} & \frac{\partial^2 f(x)}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f(x)}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f(x)}{\partial x_2 \partial x_1} & \frac{\partial^2 f(x)}{\partial x_2^2} & \cdots & \frac{\partial^2 f(x)}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f(x)}{\partial x_n \partial x_1} & \frac{\partial^2 f(x)}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f(x)}{\partial x_n^2} \end{bmatrix}.$$

3.1. Write the gradient and Hessian matrix for the linear function

$$f(x) = 2a^T x,$$

where $x \in \mathbb{R}^n$ and vector $a \in \mathbb{R}^n$. (2 pts)

$$\mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$f(\mathbf{x}) = 2\mathbf{a}^\top \mathbf{x} = \sum_{i=1}^{n} 2a_i x_i$$

$$\frac{\partial f(\mathbf{x})}{\partial x_j} = 2a_j$$

$$\implies \nabla_{\mathbf{x}} f = \begin{bmatrix} 2a_1 \\ 2a_2 \\ \vdots \\ 2a_n \end{bmatrix}$$

$$\nabla_{\mathbf{x}} f = 2\mathbf{a}$$

$\nabla_{\mathbf{x}} f(x) \in \mathbb{R}^n$

$$\nabla^2_{\mathbf{x}} f(x) = 0_{n\times n}$$

$\nabla^2_{\mathbf{x}} f(x)$ is the 0 matrix of size $n \times n$

3.2. Write the gradient and Hessian matrix of the quadratic function

$$f(x) = \frac{1}{2} x^T A x - b^T x + c,$$

where $x \in \mathbb{R}^n$, matrix $A \in \mathbb{R}^{n\times n}$, vector $b \in \mathbb{R}^n$, and $c \in \mathbb{R}$. (2 pts)
Individually taking the gradient of each term in $f(x)$ Let $f_1(x) = \frac{1}{2} x^T A x$

$$\nabla_{\mathbf{x}} f_1(\mathbf{x}) = \left(\frac{df_1(\mathbf{x})}{d\mathbf{x}}\right)^\top$$

$$\frac{df_1(\mathbf{x})}{\mathbf{x}} = \frac{1}{2} \times (x^T \frac{d(\mathbf{A}\mathbf{x})}{\mathbf{x}} + x^T A^T \frac{d(\mathbf{x})}{\mathbf{x}})$$

$$\frac{df_1(\mathbf{x})}{\mathbf{x}} = \frac{1}{2} \times (x^T A + x^T A^T)$$

$$\frac{df_1(\mathbf{x})}{\mathbf{x}} = \frac{1}{2} \times (x^T (A + A^T))$$

$$\nabla_{\mathbf{x}} f_1(\mathbf{x}) = \frac{1}{2} \times (x^T (A + A^T))^T = \frac{1}{2} \times (A + A^T) x$$

Let $f_2(x) = -b^T x$

$$\nabla_{\mathbf{x}} f_2(\mathbf{x}) = -b^T$$

Let $f_3(x) = c$

$$\nabla_{\mathbf{x}} f_3(\mathbf{x}) = 0$$

$$\nabla_{\mathbf{x}} f(\mathbf{x}) = \frac{1}{2} \times (A + A^T) x - b^T$$

$\nabla_{\mathbf{x}} f(\mathbf{x})$ is a $1 \times n$ vector.

$$\nabla^2_{\mathbf{x}} f(x) = \frac{1}{2} \times (A + A^T)$$

$\nabla^2_{\mathbf{x}} f(x)$ is a $n \times n$ matrix.

If A is symmetric, $\nabla_{\mathbf{x}} f(x) = Ax - b^T$ and $\nabla^2_{\mathbf{x}} f(x) = A$

3.3. Given matrix $A \in \mathbb{R}^{m\times n}$ where $m < n$ and $\text{rank}(A) = m$, and vector $b \in \mathbb{R}^m$, find a solution $x \in \mathbb{R}^n$ such that $Ax = b$. (3 pts)

Let $A \in \mathbb{R}^{m\times n}$ with full row rank ($m \le n$). We want A^+ such that

$$AA^+ = I_m.$$

Assume

$$A^+ = A^\top X, \quad X \in \mathbb{R}^{m\times m}.$$

Substitute:

$$AA^+ = A(A^\top X) = (AA^\top)X = I_m \implies X = (AA^\top)^{-1}.$$

We get the right pseudo-inverse to be

$$A^+ = A^\top (AA^\top)^{-1}.$$

Hence, the solution to the above equation can be given by

$$x = A^\top (AA^\top)^{-1} b$$

3.4. Given a nonsingular matrix

$$M = \begin{bmatrix} A & B \\ C & D \end{bmatrix},$$

where elements $A, B, C, D \in \mathbb{R}^{2\times 2}$, write an analytic solution of M^{-1}.

a. Assume that matrix A is not singular. (2 pts)

b. Assume that matrix D is not singular. (2 pts)

c. Assume that both matrices A and D are singular. (4 pts)

Hint: Refer to Theorem 2.1 on Page 2 of the paper: *Lu, Tzon-Tzer and Sheng-Hua Shiou. "Inverses of 2 × 2 block matrices." Computers & Mathematics With Applications 43 (2002): 119-129*

Solution:

Theorem (Lu–Shiou, Theorem 2.1: square diagonal partition)

Assume A and D are square blocks.

(i) If A is nonsingular, then M is invertible if and only if the Schur complement

$$S_A := D - CA^{-1}B$$

is nonsingular, and

$$M^{-1} = \begin{bmatrix} A^{-1} + A^{-1}BS_A^{-1}CA^{-1} & -A^{-1}BS_A^{-1} \\ -S_A^{-1}CA^{-1} & S_A^{-1} \end{bmatrix}.$$

(ii) If D is nonsingular, then M is invertible if and only if the Schur complement

$$S_D := A - BD^{-1}C$$

is nonsingular, and

$$M^{-1} = \begin{bmatrix} S_D^{-1} & -S_D^{-1}BD^{-1} \\ -D^{-1}CS_D^{-1} & D^{-1} + D^{-1}CS_D^{-1}BD^{-1} \end{bmatrix}.$$

Theorem (Lu–Shiou, Theorem 2.2: square off-diagonal partition)

Assume B and C are square blocks.

(i) If B is nonsingular, then M is invertible if and only if the Schur complement

$$S_B := C - DB^{-1}A$$

is nonsingular, and

$$M^{-1} = \begin{bmatrix} -S_B^{-1}DB^{-1} & S_B^{-1} \\ B^{-1} + B^{-1}AS_B^{-1}DB^{-1} & -B^{-1}AS_B^{-1} \end{bmatrix}.$$

(ii) If C is nonsingular, then M is invertible if and only if the Schur complement

$$S_C := B - AC^{-1}D$$

is nonsingular, and

$$M^{-1} = \begin{bmatrix} -C^{-1}DS_C^{-1} & C^{-1} + C^{-1}DS_C^{-1}AC^{-1} \\ S_C^{-1} & -S_C^{-1}AC^{-1} \end{bmatrix}.$$

Solution using the theorem

Let

$$M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}, \qquad A, B, C, D \in \mathbb{R}^{2\times 2},$$

and assume M is nonsingular.

(a) Assume A is nonsingular. Apply Theorem 2.1(i) with $S_A = D - CA^{-1}B$:

$$M^{-1} = \begin{bmatrix} A^{-1} + A^{-1}BS_A^{-1}CA^{-1} & -A^{-1}BS_A^{-1} \\ -S_A^{-1}CA^{-1} & S_A^{-1} \end{bmatrix}.$$

(b) Assume D is nonsingular. Apply Theorem 2.1(ii) with $S_D = A - BD^{-1}C$:

$$M^{-1} = \begin{bmatrix} S_D^{-1} & -S_D^{-1}BD^{-1} \\ -D^{-1}CS_D^{-1} & D^{-1} + D^{-1}CS_D^{-1}BD^{-1} \end{bmatrix}.$$

(c) Assume both A and D are singular. Since Theorem 2.1 does not apply, use the off-diagonal partition (Theorem 2.2):

- If B is nonsingular, apply Theorem 2.2(i) with $S_B = C - DB^{-1}A$:

$$M^{-1} = \begin{bmatrix} -S_B^{-1}DB^{-1} & S_B^{-1} \\ B^{-1} + B^{-1}AS_B^{-1}DB^{-1} & -B^{-1}AS_B^{-1} \end{bmatrix}.$$

- If C is nonsingular, apply Theorem 2.2(ii) with $S_C = B - AC^{-1}D$:

$$M^{-1} = \begin{bmatrix} -C^{-1}DS_C^{-1} & C^{-1} + C^{-1}DS_C^{-1}AC^{-1} \\ S_C^{-1} & -S_C^{-1}AC^{-1} \end{bmatrix}.$$

3.5. Given a nonsingular matrix

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix},$$

write the analytic solution of A^{-1}. (4 pts)

Solution:

Let

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}, \qquad \det(A) \neq 0.$$

We derive A^{-1} using the adjugate formula:

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A),$$

where $\operatorname{adj}(A)$ is the transpose of the cofactor matrix of A.

Compute $\det(A)$

Expand the determinant along the first row:

$$\begin{aligned} \det(A) &= a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix} \\ &= a(ei - fh) - b(di - fg) + c(dh - eg). \end{aligned}$$

Denote

$$\Delta = \det(A) = a(ei - fh) - b(di - fg) + c(dh - eg).$$

Form the cofactor matrix

For each entry a_{pq} of A, define the *minor* M_{pq} as the determinant of the 2×2 matrix obtained by deleting row p and column q. The *cofactor* is

$$C_{pq} = (-1)^{p+q} M_{pq}.$$

Compute minors and cofactors:

Row 1 cofactors

$$M_{11} = \begin{vmatrix} e & f \\ h & i \end{vmatrix} = ei - fh, \qquad C_{11} = (+1)M_{11} = ei - fh,$$

$$M_{12} = \begin{vmatrix} d & f \\ g & i \end{vmatrix} = di - fg, \qquad C_{12} = (-1)M_{12} = -(di - fg),$$

$$M_{13} = \begin{vmatrix} d & e \\ g & h \end{vmatrix} = dh - eg, \qquad C_{13} = (+1)M_{13} = dh - eg.$$

Row 2 cofactors

$$M_{21} = \begin{vmatrix} b & c \\ h & i \end{vmatrix} = bi - ch, \qquad C_{21} = (-1)M_{21} = -(bi - ch),$$

$$M_{22} = \begin{vmatrix} a & c \\ g & i \end{vmatrix} = ai - cg, \qquad C_{22} = (+1)M_{22} = ai - cg,$$

$$M_{23} = \begin{vmatrix} a & b \\ g & h \end{vmatrix} = ah - bg, \qquad C_{23} = (-1)M_{23} = -(ah - bg).$$

Row 3 cofactors

$$M_{31} = \begin{vmatrix} b & c \\ e & f \end{vmatrix} = bf - ce, \qquad C_{31} = (+1)M_{31} = bf - ce,$$

$$M_{32} = \begin{vmatrix} a & c \\ d & f \end{vmatrix} = af - cd, \qquad C_{32} = (-1)M_{32} = -(af - cd),$$

$$M_{33} = \begin{vmatrix} a & b \\ d & e \end{vmatrix} = ae - bd, \qquad C_{33} = (+1)M_{33} = ae - bd.$$

Thus the cofactor matrix $C = [C_{pq}]$ is

$$C = \begin{bmatrix} (ei - fh) & -(di - fg) & (dh - eg) \\ -(bi - ch) & (ai - cg) & -(ah - bg) \\ (bf - ce) & -(af - cd) & (ae - bd) \end{bmatrix}.$$

Take the adjugate

By definition,

$$\operatorname{adj}(A) = C^T.$$

Therefore,

$$\operatorname{adj}(A) = \begin{bmatrix} (ei - fh) & -(bi - ch) & (bf - ce) \\ -(di - fg) & (ai - cg) & -(af - cd) \\ (dh - eg) & -(ah - bg) & (ae - bd) \end{bmatrix}.$$

Apply the adjugate formula

Finally,

$$A^{-1} = \frac{1}{\Delta}\operatorname{adj}(A) = \frac{1}{a(ei-fh)-b(di-fg)+c(dh-eg)}\begin{bmatrix} (ei-fh) & -(bi-ch) & (bf-ce) \\ -(di-fg) & (ai-cg) & -(af-cd) \\ (dh-eg) & -(ah-bg) & (ae-bd) \end{bmatrix}.$$

3.6. Consider a matrix $A = [a_{i,j}] \in \mathbb{R}^{2\times 2}$ which is **not singular** Derive the analytical form of the derivative of f over matrix A (i.e. $[u_{i,j}] = \nabla_A f$, where $u_{i,j} = \partial f/\partial a_{i,j}$) for function $f = \operatorname{tr} A^{-1}$. (5 pts)

Solution:

Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathbb{R}^{2\times 2}, \quad \det(A) \neq 0,$$

$$A^{-1} = \frac{1}{\det(A)}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}, \quad \det(A) = ad - bc.$$

$$f(A) = \operatorname{tr}(A^{-1}) = \frac{a+d}{ad-bc}.$$

$$\frac{\partial f}{\partial a} = \frac{(1)(ad-bc)-(a+d)(d)}{(ad-bc)^2} = \frac{-d^2-bc}{(ad-bc)^2}$$

$$\frac{\partial f}{\partial b} = \frac{0\cdot(ad-bc)-(a+d)(-c)}{(ad-bc)^2} = \frac{c(a+d)}{(ad-bc)^2}$$

$$\frac{\partial f}{\partial c} = \frac{0\cdot(ad-bc)-(a+d)(-b)}{(ad-bc)^2} = \frac{b(a+d)}{(ad-bc)^2}$$

$$\frac{\partial f}{\partial d} = \frac{1\cdot(ad-bc)-(a+d)(a)}{(ad-bc)^2} = \frac{-a^2-bc}{(ad-bc)^2}$$

$$\nabla_A f = \begin{bmatrix} \frac{\partial f}{\partial a} & \frac{\partial f}{\partial b} \\ \frac{\partial f}{\partial c} & \frac{\partial f}{\partial d} \end{bmatrix} = \frac{1}{(ad-bc)^2}\begin{bmatrix} -(d^2+bc) & c(a+d) \\ b(a+d) & -(a^2+bc) \end{bmatrix}.$$